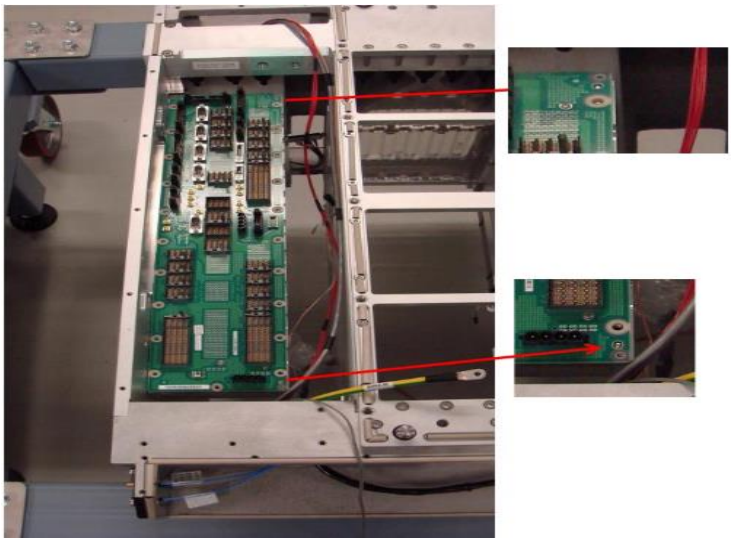
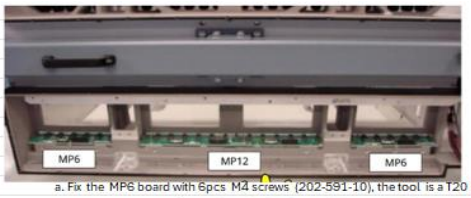
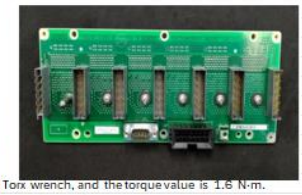
Build Overview:

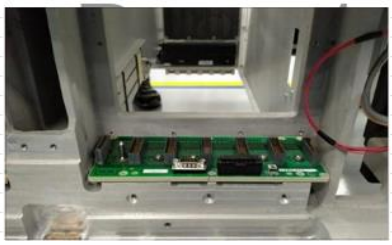
1 **Support Board Backplane Install**
2 **LED/EMO and Ground Cable Install**
3 **NVRAM PCB**
4 **Install Mid-Planes**
5 **Elbow Tube Fitting Install**
6 **Pressure Sensor Install**
7 **Install the following cables:**
- **a.SB24 48V Power Cable**
- **b.EMO**
- **c.DSI Power**
- **d.IEC Power**
- **e.Scope Panel**
- **f.TDR Panel**







a. Fix the MP6 board with 6pcs M4 screws (202-591-10), the tool is a T20 Torx wrench, and the torque value is 1.6 N-m.



CHANGE HISTORY

Date	BPM Rev	Change Details	Updated by
	1	New Issue	JOSEPH AMAD

Prepared By:
JOSEPH R. AMAD
Manufacturing Engineer

Checked By:
IVAN NOWELL PANIQUE
Team Lead

Approved By:
SHERWIN ARIOSTE
Operations Manager

MAN: CERTIFIED TECHNICIANS.

MATERIAL:

NO.	PART NO.	DESCRIPTION	QTY.
1	650-705-00	CABLE, UF2 12SLOT SB25 INPUT POWER, TELECOM, WITH CONN	1
2	629-576-00	CABLE DSI POWER SPLIT INTERNAL TH TELECOM WITH CONN	1
3	710-502-55	CBL/ASSY 6-FT. HEAVY-DUTY 14AWG POWER CORD (IEC-320-C13 TO IEC-320-C14)	2
4	360-174-01	VELCRO, 10PCS 1/4" X 5"VELCRO BRAND ONE-WRAP TIES 10 CT- 91594, PURPLE	2
5	654-656-01	ELEC-MECH, UFII SP6 SIDEPLANE BOARD ASSY, FOR SEMI TEST	1
6	472-426-16	PAN HEAD CROSS RECESSED SCREW M5X0.8 x 12MM L W/CONE WASHER, O.D, 11MM	22
7	835-610-00	HDW CLAMP CBL .312	3
8	813-929-01	HDW VELCRO STRP 5/8	4
9	202-591-06	SCRW PAN HEAD TORX M4X.7P/6MM SS W/NYLOK	1
10	634-157-00	PCBA, ULTRAFLEX II SYS NVRAM, SEMI TEST	1
11	629-403-01	ELEC-MECH, UFII LARGE MIDPLANE ASSY, FOR SEMI TEST	1
12	202-591-10	SCRW PAN HEAD TORX M4X.7P/10MM SS W/NYLOK	22
13	629-401-02	ELEC-MECH, UFII SMALL MIDPLANE ASSY, FOR SEMI TEST	2
14	361-754-05	FIT ELBW 3/8" TUBE ODX9/16-18 MALE SAE/MS SS W/EPDM O-RING	1
15	638-030-00	DOW CORNING MOLYKOTE 111 COMPOUND GRAY 6G PACKGE	1
16	638-079-01	PRESSURE SENSOR 0-100PSIG 4-20MA SAE #4 PORT 7/16"-20 UNF WITHOUT 2 O-RING	2
17	361-607-08	O-RING 0.351 "IDX 0.072"W E0540-80	2
18	638-309-02	ALUMINUM, PANEL, DISCONNECT, BRACKET OR MISC ARTICLE	1
19	406-502-10	SCRW PAN HEAD TORX M5X.8P/10MM SS W/NYLOK	6

METHOD:

Flow chart of operation steps

BPM preparation
↓
Preparation Tools
↓
Prepare Materials
↓
Complete Operation
↓
Inventory Materials
↓
Operation Check
↓
Inventory tool

MACHINE:

TERADYNE

TOOL NAME	TORQUE VALUE
FLAT-BLADE SCREWDRIVER	0.7 N.m
PH2 PHILLIPS BIT SCREWDRIVER	3.3 N.m
PH1 PHILLIPS HEAD	0.7 N.m
OPEN-END WRENCH	27 N.m
PH2 CROSS CUTTER HEAD	1.6 N.m
PH2 PHILLIPS SCREWDRIVER	2.8 N.m
22MM LONG SOCKET WRENCH	27 N.m
T25 TORX WRENCH	3.3 N.m
HEX 3MM TORX WRENCH	1.6 N.m
T20 TORX WRENCH	1.6 N.m
3/16 SOCKET WRENCH	0.19 N.m
16MM OPEN END	N/A

MAN: CERTIFIED OPERATOR, TECHNICIAN

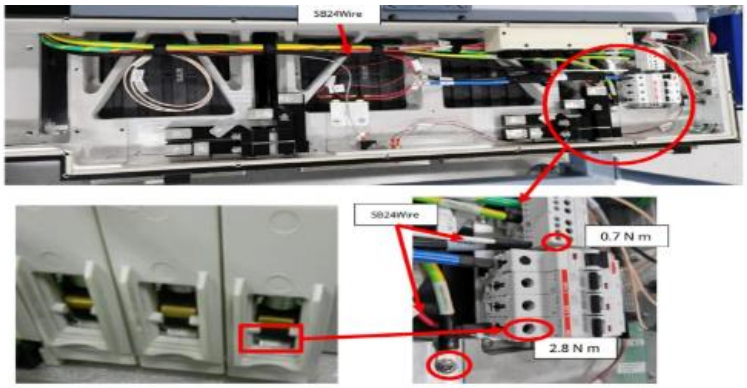
MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
1	650-705-00	CABLE, UF2 12SLOT SB25 INPUT POWER, TELECOM, WITH CONN	1

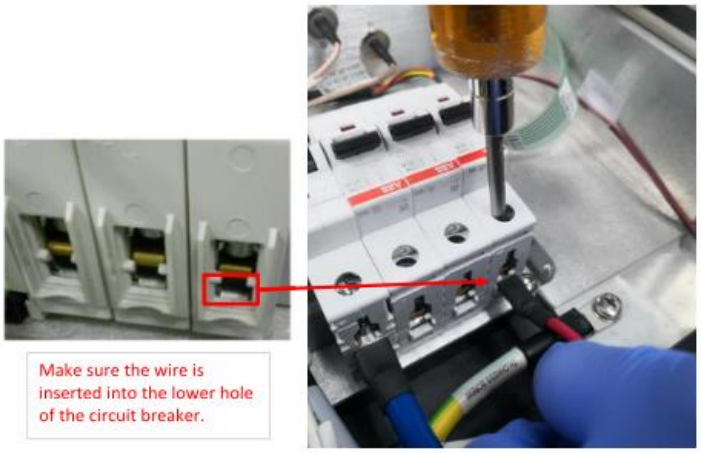
MACHINE: flat-blade screwdriver 0.7 N.m, PH2 Phillips bit screw driver 2.8 N.m.

METHOD:

1.1 Install a SB24 48V line (650-705-00), and place the SB24 line along the EMO line to the circuit breaker position as shown in the figure.



Insert the white wire and use a flat-blade screwdriver with a tool, andthetorque value is0.7N·m.



Make sure the wire is inserted into the lower hole of the circuit breaker.

Insert the red wire and use the tool PH2 Phillips bit screwdriver with a torquevalue of2.8 N·m.

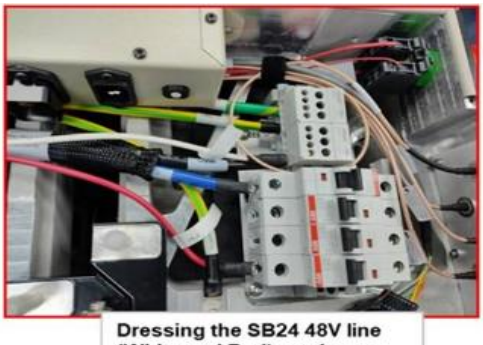
MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

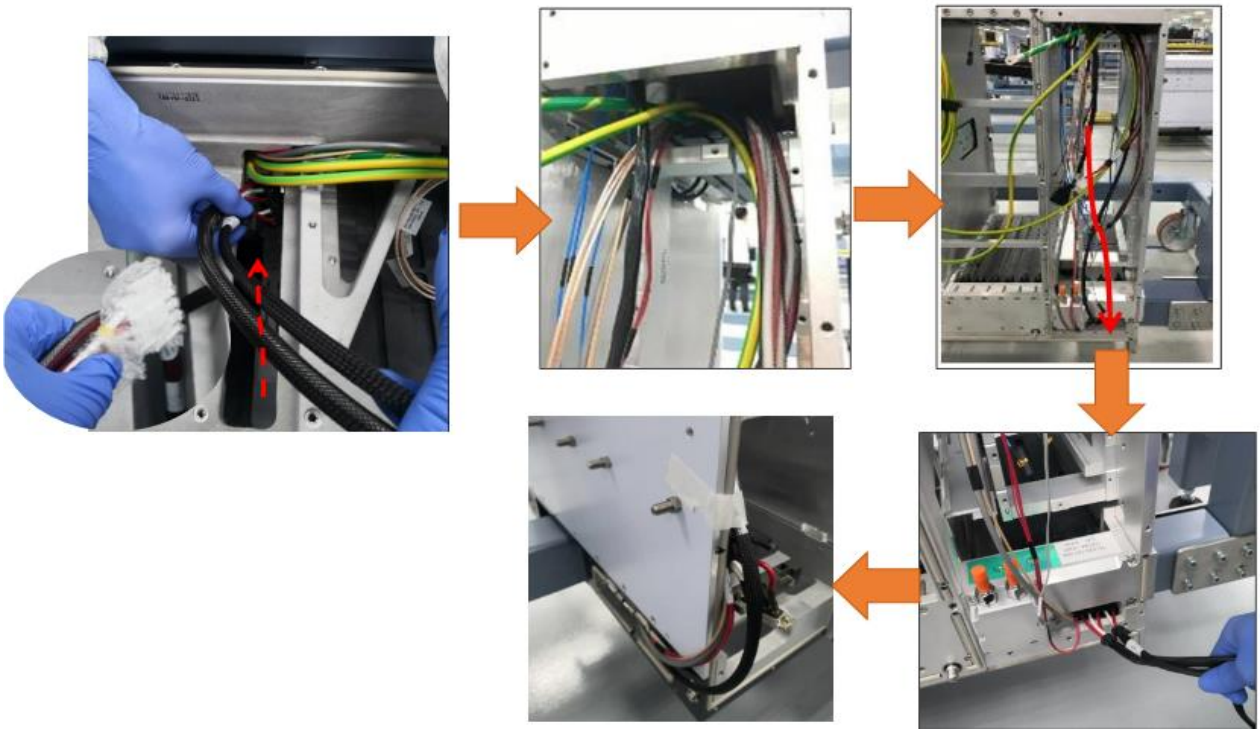
METHOD:

1.5 Bundle the SB24 wire and all cable with the exsiting velcro tie.



Dressing the SB24 48V line (White and Red) as shown.

1.6 Pass the switch wire into the small hole along the T/H frame as shown in the figure, pass through the channel to the backplane, and tie the wire tightly using the existing cable tie.



Safety Tips: N/A

TQC: N/A

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MAN: CERTIFIED OPERATOR, TECHNICIAN

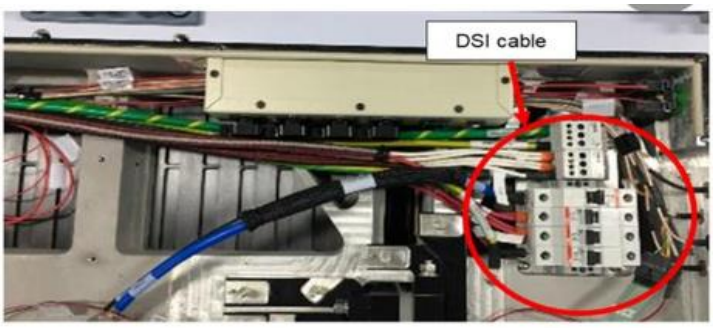
MATERIAL:

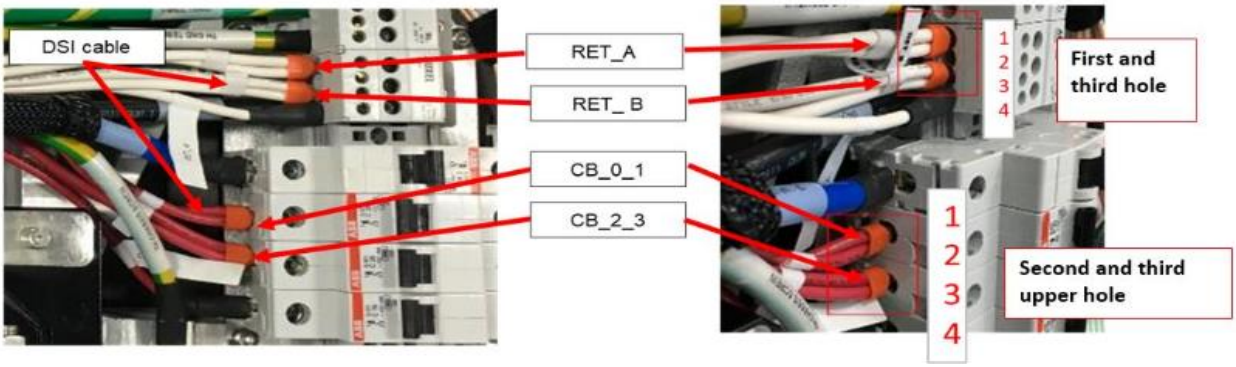
ITEM	PART NUMBER	DESCRIPTION	QTY
2	629-576-00	CABLE DSI POWER SPLIT INTERNAL TH TELECOM WITH CONN	1

MACHINE: flat-blade screwdriver 0.7 N.m, PH2 Phillips bit screw driver 2.8 N.m.

METHOD:

2.1 Install the DSI power cord (629-576-00).





a. Insert the DSI white wire into the first and third holes in the upper row of the lower power distribution block, use a flat-blade screwdriver, and the torque value is 0.7 N·m.

b. Insert the DSI red wire into the second and third holes in the upper row of the circuit breaker as shown in the figure, and use a PH2 Phillips screwdriver with a torque of 2.8 N·m.

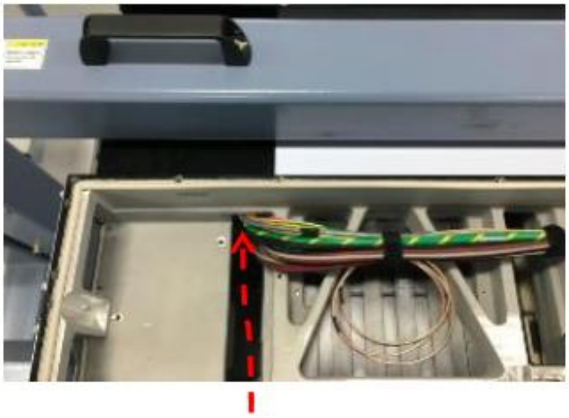
MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

c. Pass the DSI power cable into the small hole along the T/H frame as shown in the figure, pass through the channel to the backplane surface, and use the existing cable tie to temporarily tie the cable,
You will need to readjust when inserting this cable to the backplane later.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

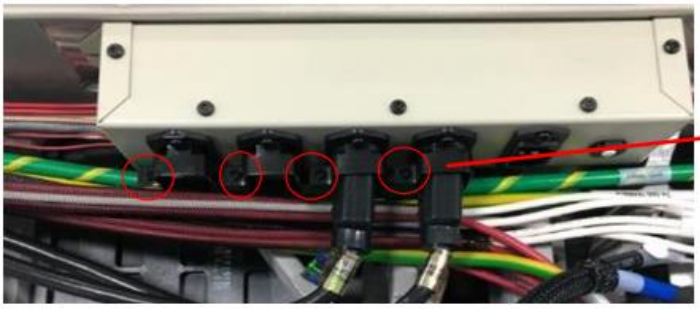
ITEM	PART NUMBER	DESCRIPTION	QTY
3	710-502-55	CBL/ASSY 6-FT. HEAVY-DUTY 14AWG POWER CORD (IEC-320-C13 TO 2 IEC-320-C14)	1

MACHINE: flat-blade screwdriver 0.7 N.m

METHOD:

3.1 Install 2pcs IEC wires (710-502-55).

a. Insert the wire into the power outlet as shown.



b. Use PH1 Phillips head to fix the four connector screws, the torque value is 0.7 N·m.

Note: Ensure that all plugs are fully inserted into the power outlet.

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MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

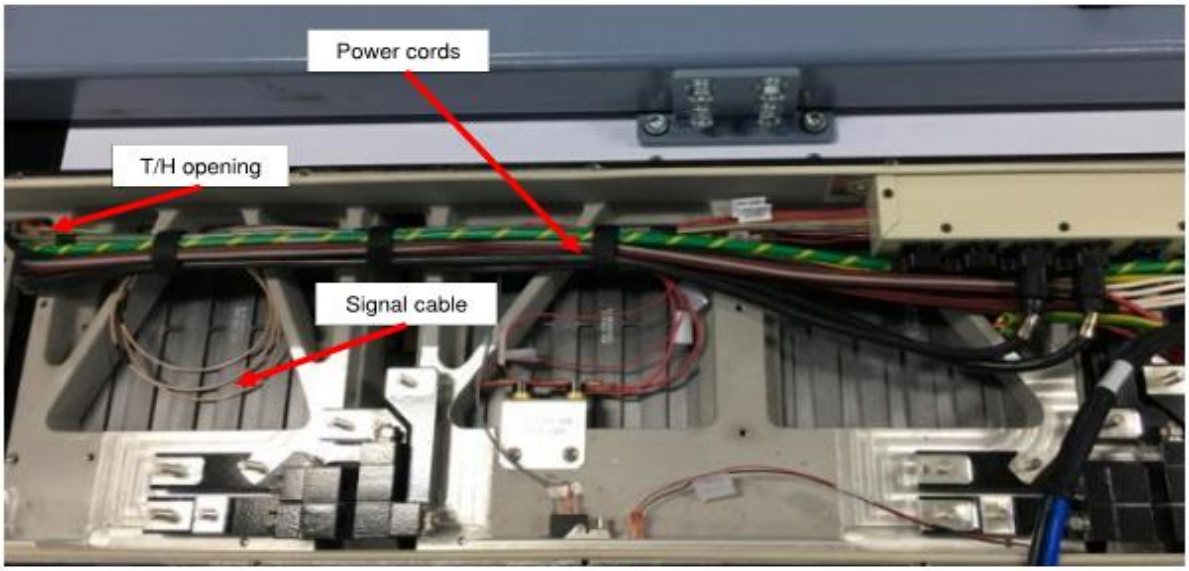
MACHINE: N/A

METHOD:

c. Place the 2 power cords along the wire harness as shown in the figure and tie them with the existing cable ties.

d. Pass the 2 power cords through the T/H opening and lead to the back panel as shown in the figure.

e. Wrap the signal wire as shown in the figure and tie it with the existing cable ties,and place it in the T/H position as shown in the figure. Make sure you reserve enough margin for the long lineInsert the cable connector into the backplane.



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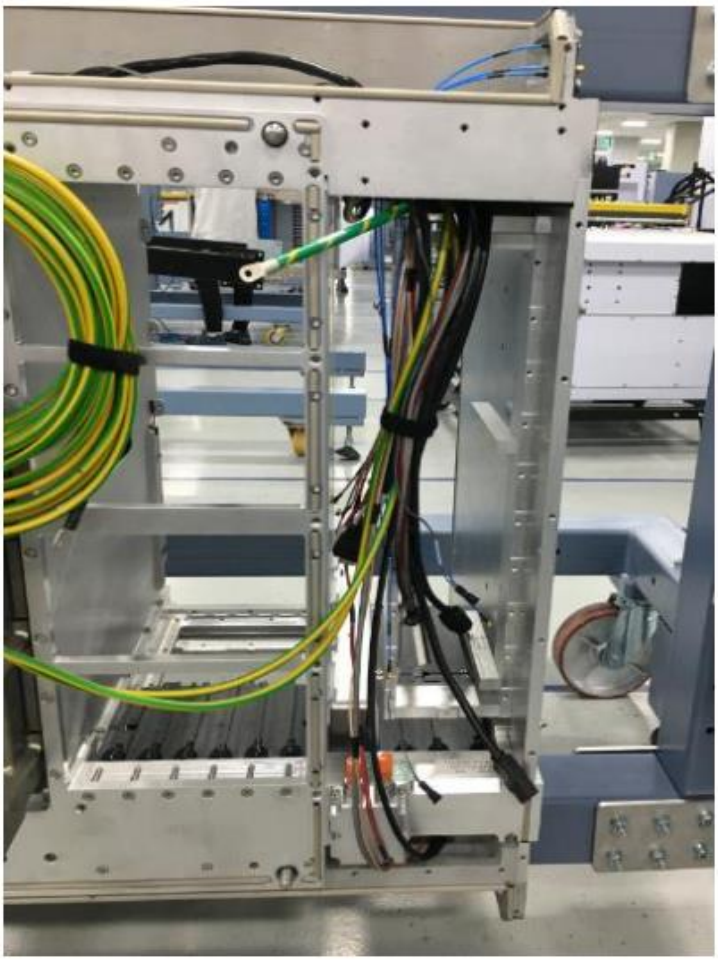
MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

f. Place the power cord on the back panel as shown in the figure, and tie it with the existing cable tie.



MAN: CERTIFIED OPERATOR, TECHNICIAN

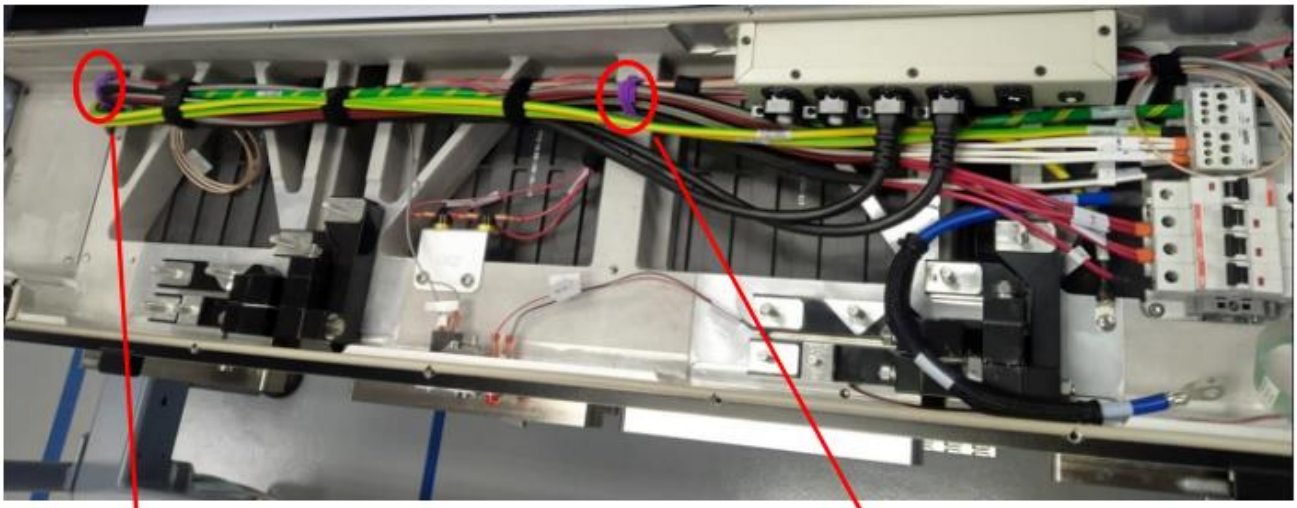
MATERIAL:

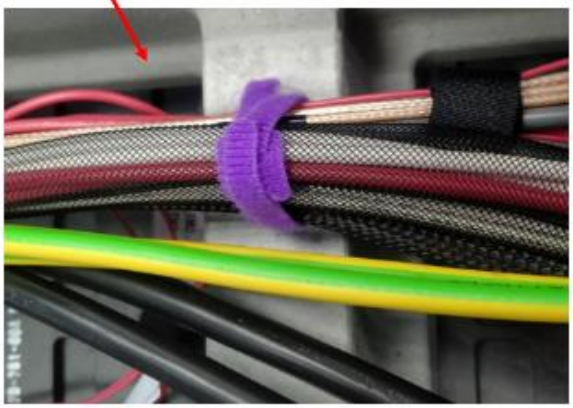
ITEM	PART NUMBER	DESCRIPTION	QTY
4	360-174-01	VELCRO, 10PCS 1/4" X 5"VELCRO BRAND ONE-WRAP TIES 10 CT-91594, PURPLE	1

MACHINE: N/A

METHOD:

g. Tighten the cable with 2 cable ties (360-174-01) as shown in the figure.





MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: Hex 3mm bit

METHOD:

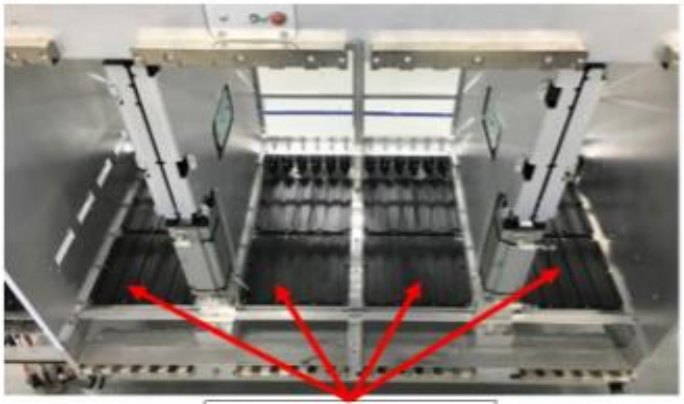
4.1 Remove the 4 plastic plates above the T/H indicated by the arrow as shown in the figure.

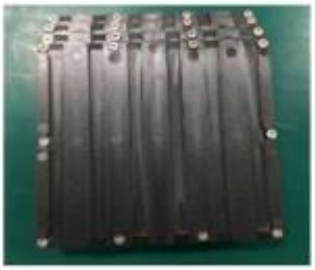
a. Use a Hex 3mm wrench to remove the short screws on the cover plate. Each cover plate has 2 short screws, total of 8 short screws.

4.2 Save the removed 4 plastic plates and the corresponding 40 screws for subsequent assembly and use, as shown in the figure, you can keep the screws in the corresponding holes.

If the screw is accidentally dropped, you can reposition it according to the screw illustration method.

Pay attention to the positions of the 8 short screws and 32 long screws, do not confuse them.





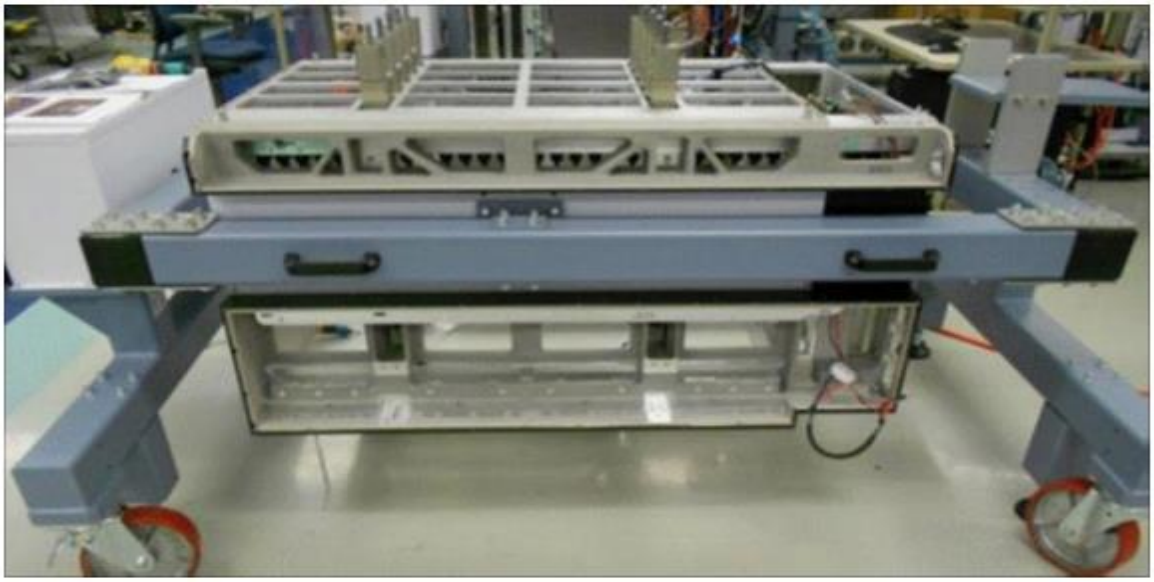
MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

5.1 Rotate the T/H direction as shown in the figure.



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MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

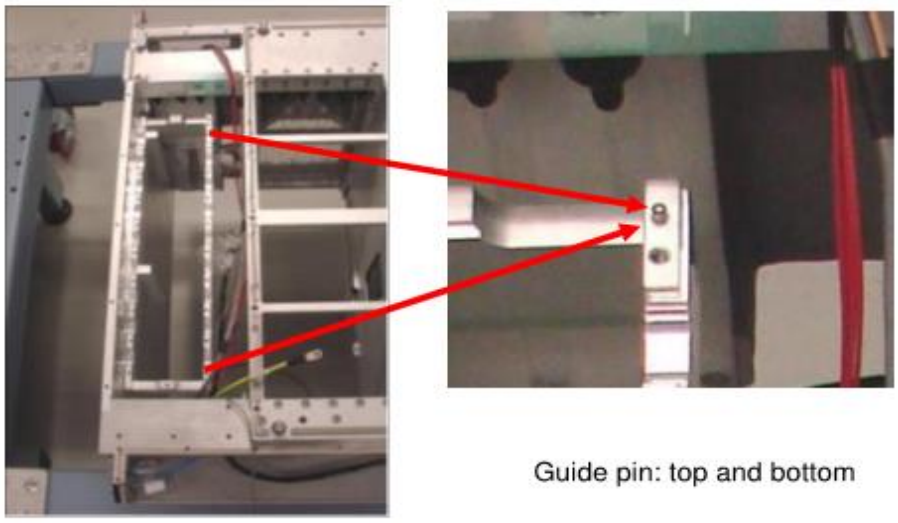
ITEM	PART NUMBER	DESCRIPTION	QTY
5	654-656-01	ELEC-MECH, UFII SP6 SIDEPLANE BOARD ASSY, FOR SEMI TEST	1

MACHINE: N/A

METHOD:

6.1 Take the backplane (654-656-01) and carefully remove it from the box. When unpacking and removing the backplane, ESD precautions must be taken.

6.2 Rotate the T/H direction as shown in the figure, and place the mounting surface of the back plate upward.
Note that the two positioning pins on the frame are used to fix the backplane.



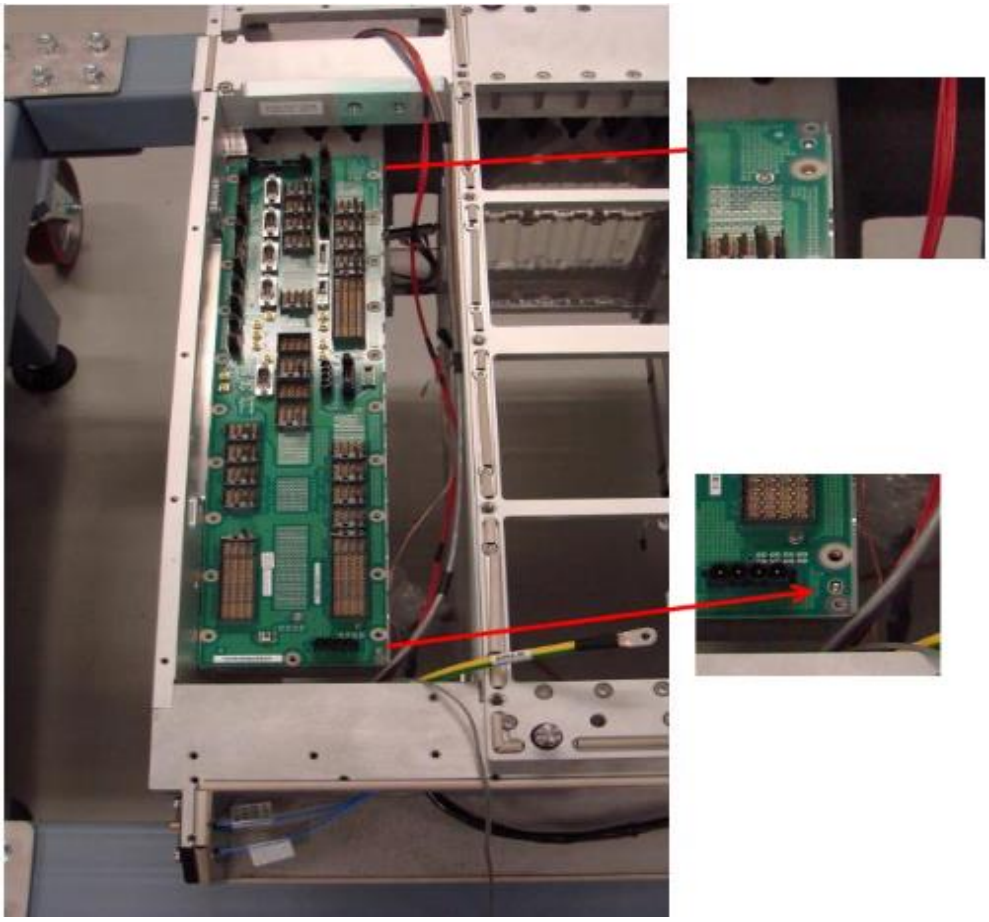
MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

6.3 Place the backplane into the T/H frame. Note that the two positioning pins in the frame must be inserted into the two positioning holes of the backplane.



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MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

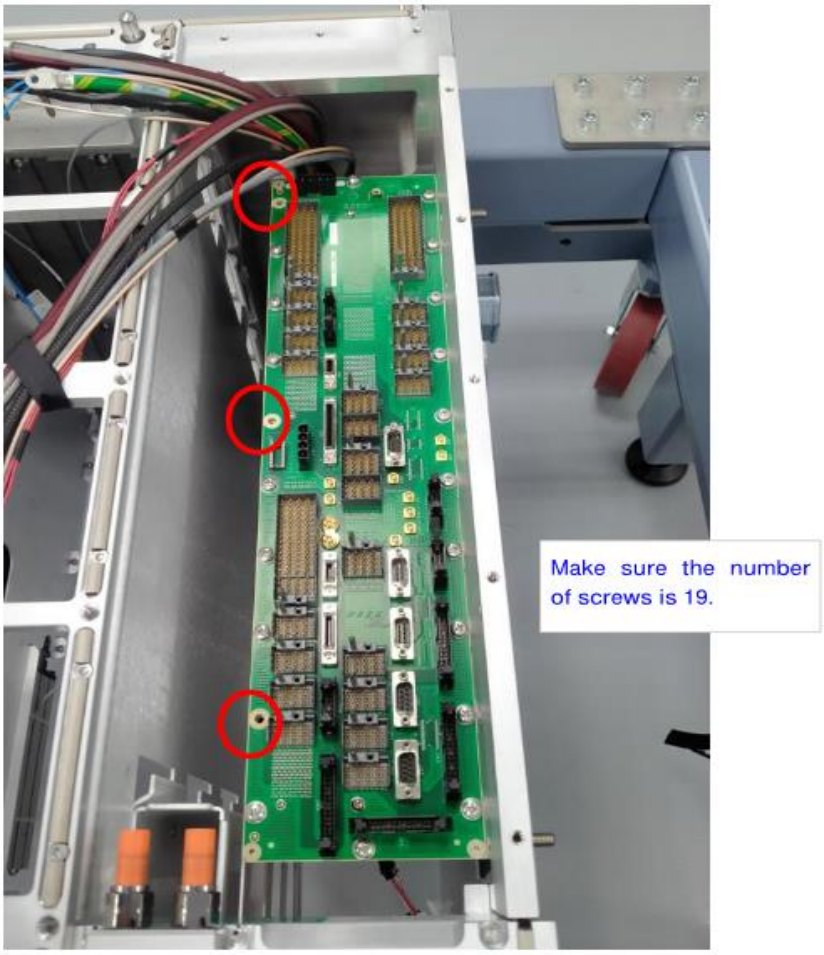
ITEM	PART NUMBER	DESCRIPTION	QTY
6	472-426-16	DESCRIPTION QTYPAN HEAD CROSS RECESSED SCREW M5X0.8x16MM L W/ CONE WASHER.22 O.D. 11MM	1

MACHINE: PH2 cross cutter head 1.6 N·m.

METHOD:

6.4 Fix the back plate on the T/H frame with 19pcs M5 screws (472-426-16), the tool is PH2 cross cutter head, and the torque value is 1.6 N·m.

Note: Do not install screws in the three red circles in the figure below.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

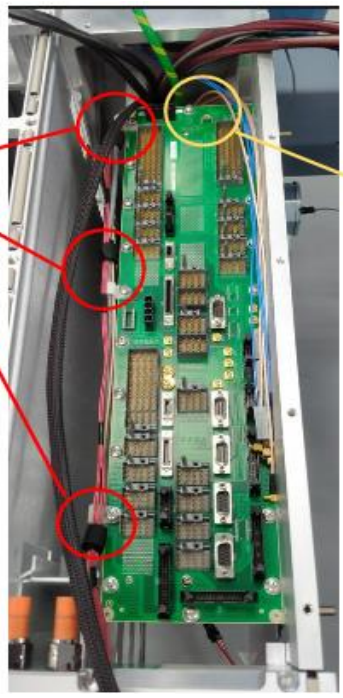
ITEM	PART NUMBER	DESCRIPTION	QTY
7	835-610-00	HDW CLAMP CBL .312	3
8	813-929-01	HDW VELCRO STRP 5/8	3
9	202-591-06	SCRW PAN HEAD TORX M4X.7P/6MM SS W/NYLOK	1

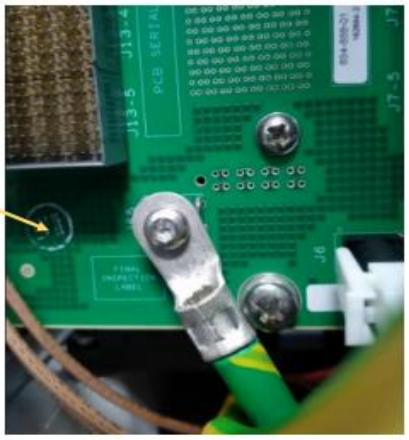
MACHINE: N/A

METHOD:

7.1 Place the EMO gray wire and 10 G HZ wire along the edge of the back panel, install 3 wire clips (835-610-00) to clamp the wire, and use 3 M5 screws (472-426-16) to fix the wire clip, Use PH2 cross cutter head with a torque value of 1.6 N m.

7.2 After locking the 3 screws, tie all the cables with 3 velcro (813-929-01).





7.3 Install the ground wire and fix it with M4 screws (202-591-06), the tool is a T20 torx wrench, and the torque value is 1.6 N·m.

TERADYNE PHILIPPINES LTD. _ CEBU

MAN: CERTIFIED OPERATOR, TECHNICIAN

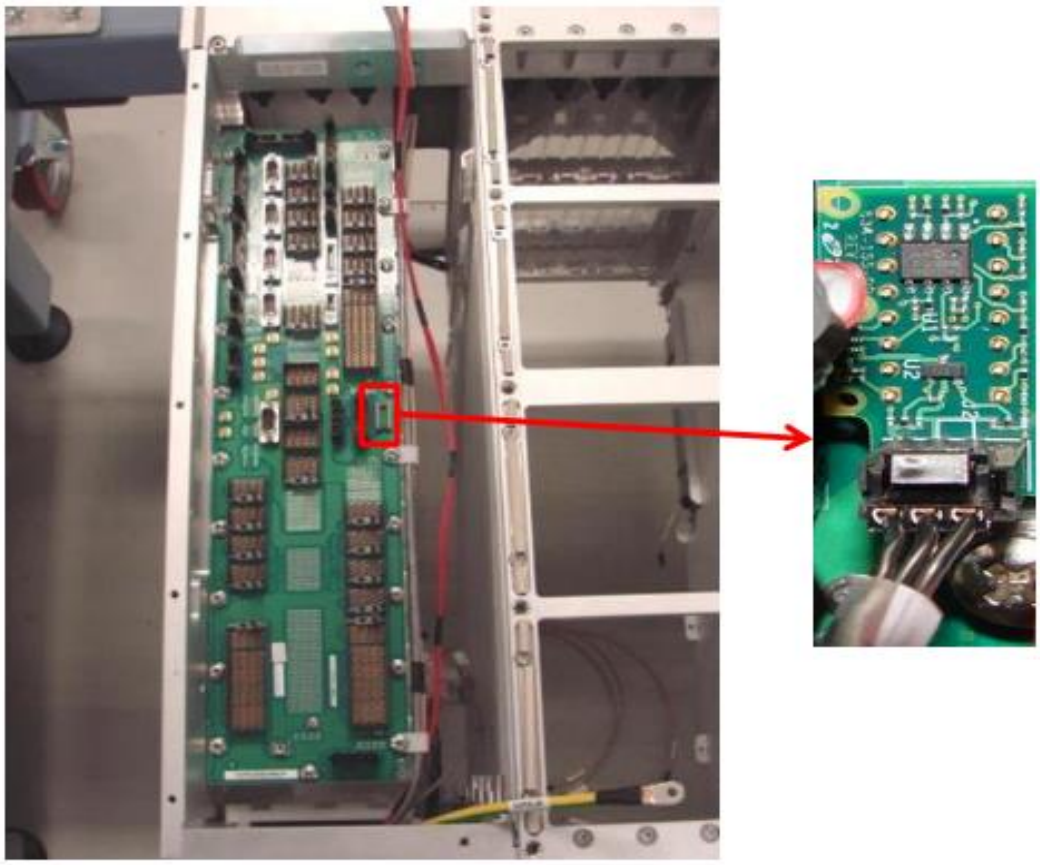
MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
10	634-157-00	PCBA, ULTRAFLEX II SYS NVRAM, SEMI TEST	1

MACHINE: N/A

METHOD:

7.4 Using ESD protection measures, install a PCBA (634-157-00) on the backplane end "J60".
The installation direction of PCBA is shown in the figure.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

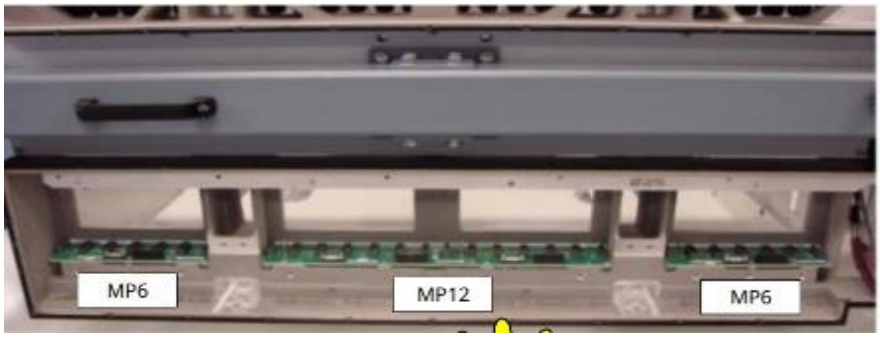
ITEM	PART NUMBER	DESCRIPTION	QTY
11	629-403-01	ELEC-MECH, UFII LARGE MIDPLANE ASSY, FOR SEMI TEST	1
12	202-591-10	SCRW PAN HEAD TORX M4X.7P/10 MM SS W/NYLOK	10

MACHINE: T20 Torx wrench, 1.6 N·m.

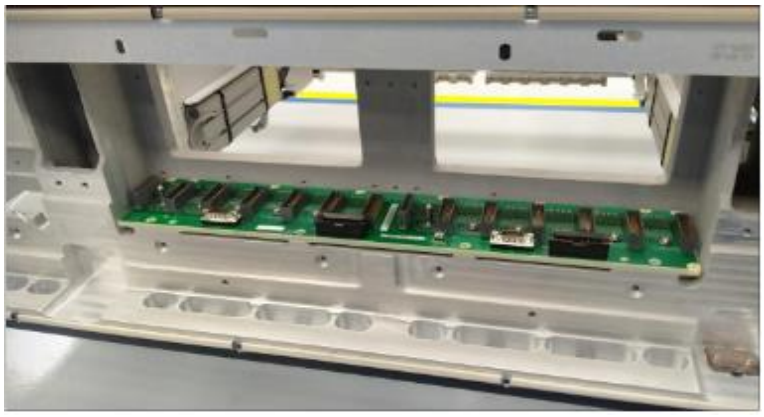
METHOD:

8.1 Install one MP12 Mid-Plane

Record the serial number in the tracker.



Fix the MP12 board with 10pcs M4 screws (202-591-10), the tool is a T20 Torx wrench, and the torque value is 1.6 N·m.



Safety Tips: N/A

TERADYNE PHILIPPINES LTD. _ CEBU

MAN: CERTIFIED OPERATOR, TECHNICIAN

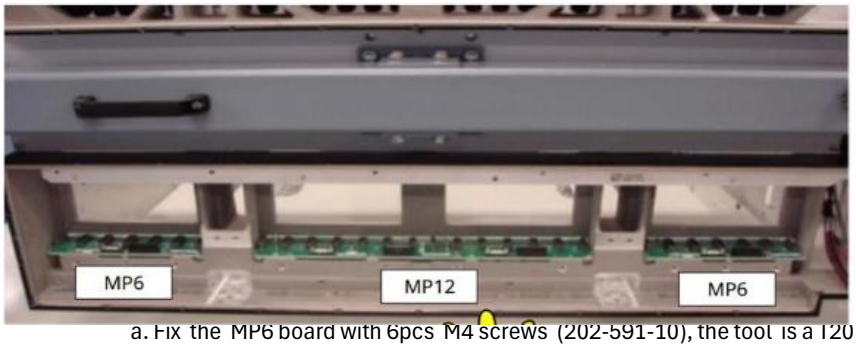
MATERIAL:

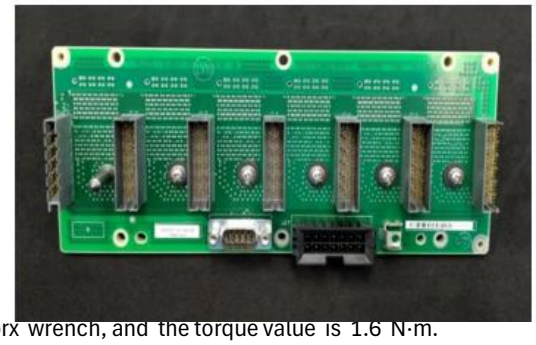
ITEM	PART NUMBER	DESCRIPTION	QTY
13	629-403-02	ELEC-MECH, UFII SMALL MIDPLANE ASSY, FOR SEMI TEST	2
14	202-591-10	SCRW PAN HEAD TORX M4X.7P/10 MM SS W/NYLOK	12

MACHINE: T20 Torx wrench, 1.6 N·m.

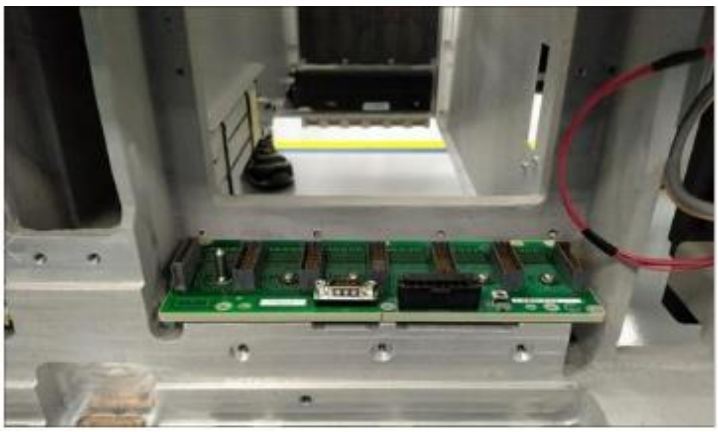
METHOD:

8.2 Install 2pcs MP6 Mid-Plane (629-401-02). Record the serial number into tracker.





a. Fix the MP6 board with 6pcs M4 screws (202-591-10), the tool is a T20 Torx wrench, and the torque value is 1.6 N·m.



d. Repeat the above actions to install another MP6 board.

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MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

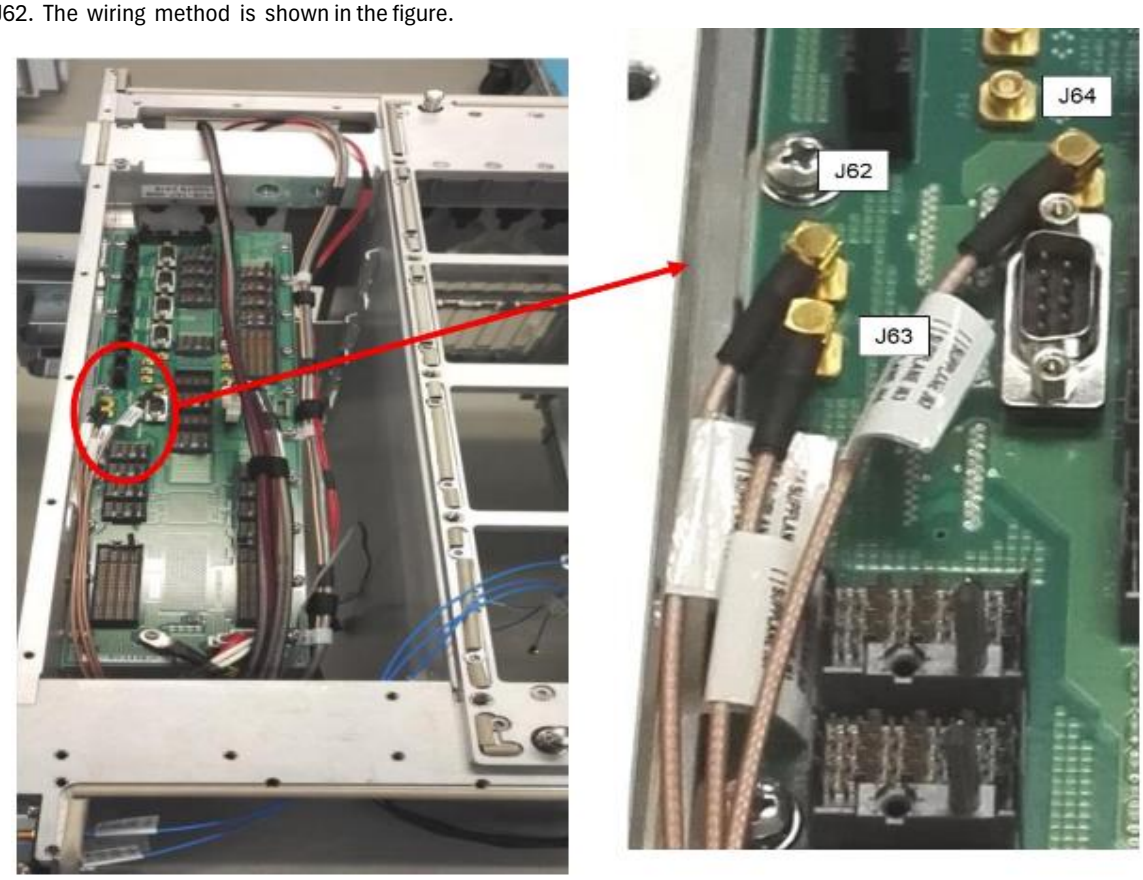
METHOD:

8.3 Install 3 panel cable connectors (627-090-00) to the backplane.

a. Rotate the T/H direction as shown, and place the backplane face up.

b. Insert the corresponding backplanes according to the online labels "SUPPLANE J64", "SUPPLANE J63" and SUPPLANE J62 as shown in the figure.

J64, J63 and J62. The wiring method is shown in the figure.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

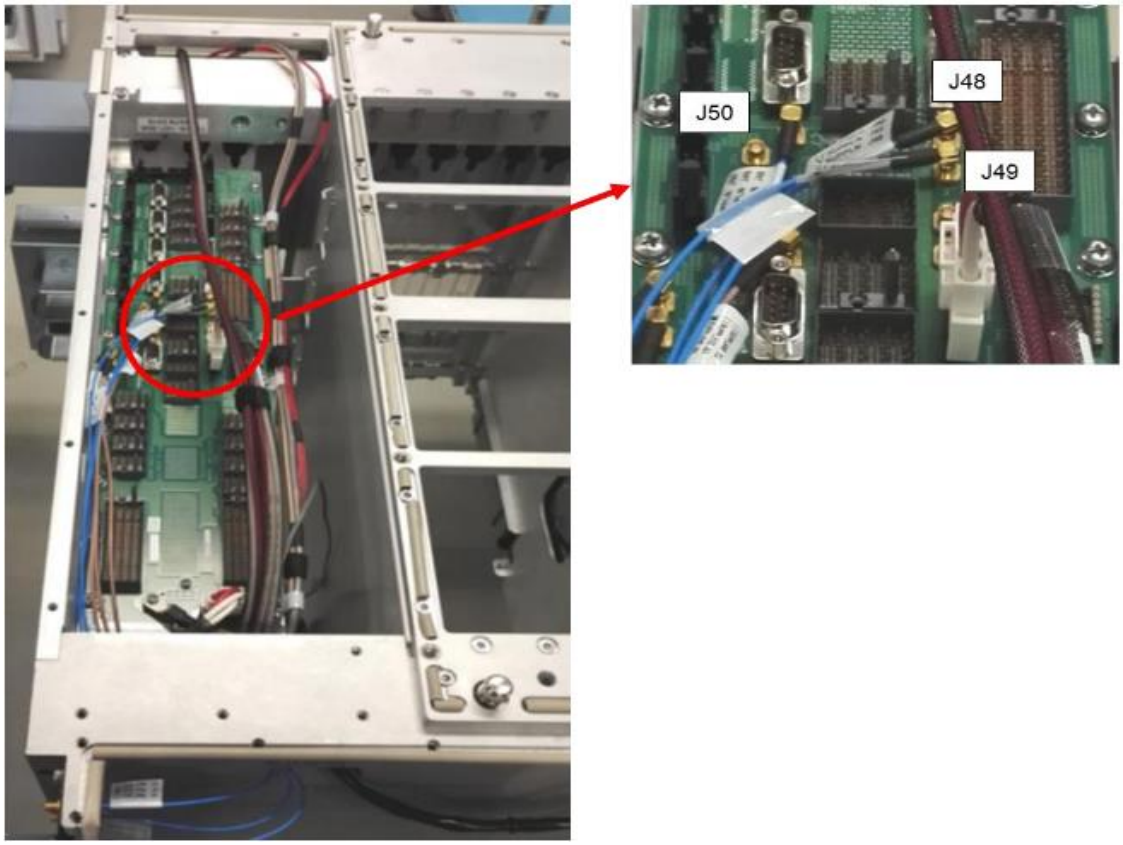
MACHINE: N/A

METHOD:

8.4 Install 3 TDR panel cable connectors (627-084-01) to the back panel.

a. Place the TDR line along the edge frame of the backplane on the backplane J48, J49, J50 area as shown in the figure.

b. Insert the corresponding backplane according to the labels "SUPPLANE J48", "SUPPLANE J49" and SUPPLANE J50 on the line as shown in the figure
J48, J49 and J50. The wiring method is shown in the figure.



TERADYNE PHILIPPINES LTD. _ CEBU

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

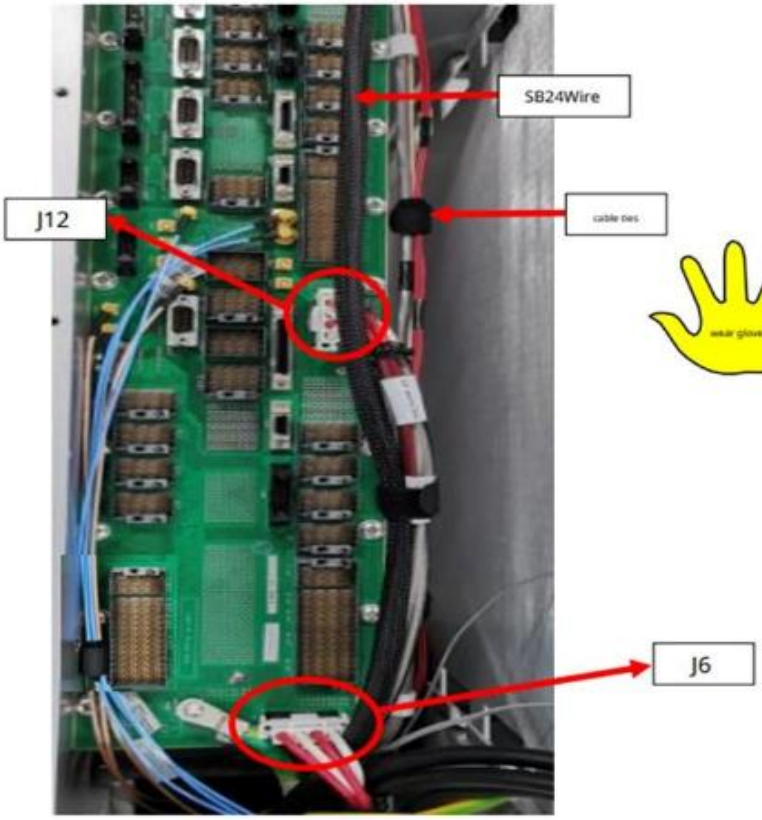
ITEM	PART NUMBER	DESCRIPTION	QTY
15	813-929-01	HDW VELCRO STRP 5/8	1

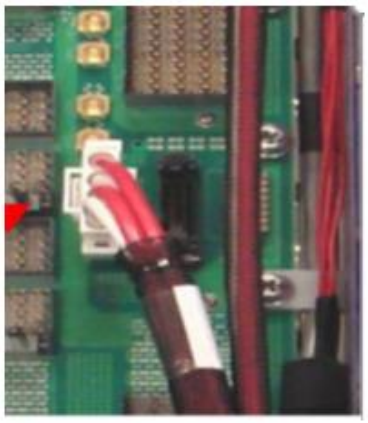
MACHINE: N/A

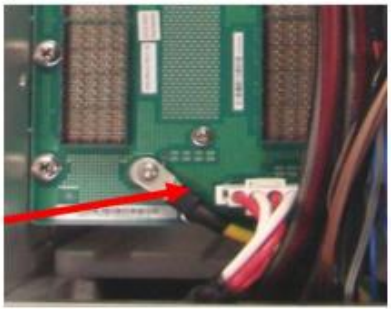
METHOD:

c. Insert the DSI power cord connectors J12 and J6 on the backplane as shown in the figure.

d. Tie the DSI power cable connectors J12 and J48 (SB24) together with cable ties (813-929-01) as shown.







TERADYNE PHILIPPINES LTD. _ CEBU

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

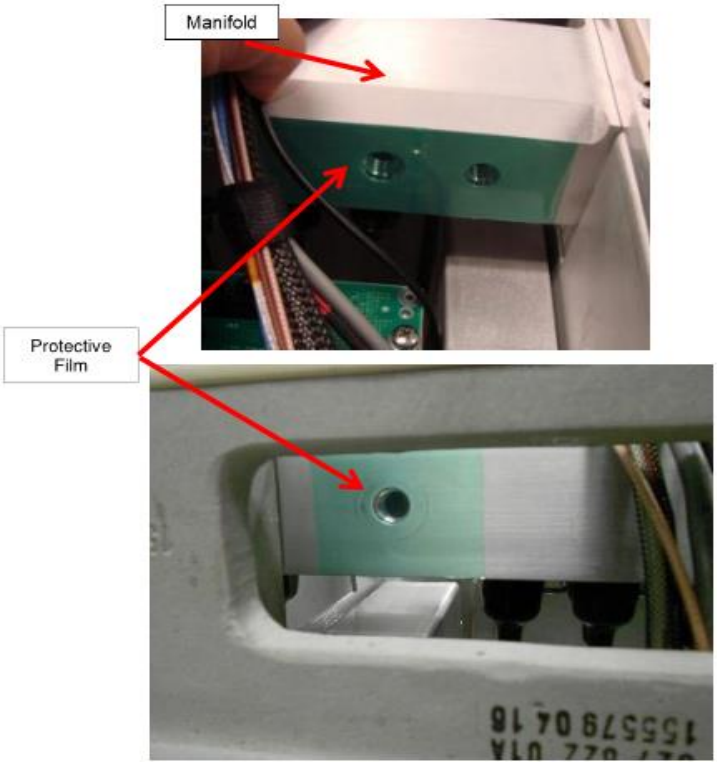
ITEM	PART NUMBER	DESCRIPTION	QTY
16	361-754-05	FIT ELBW 3/8" TUBE ODX9/16-18 MALE SAE/MS SS W/EPDM O-RING	1

MACHINE: N/A

METHOD:

9.1 Install 1pcs 3/8"Pipe connector (361-754-05).

a. Carefully remove the protective film under the Manifold and check that the Manifold inlet and pressure sensor port are not damaged or broken.



TERADYNE PHILIPPINES LTD. _ CEBU

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
17	638-030-00	DOW CORNING MOLYKOTE 111 COMPOUND GRAY 6G PACKAGE	1

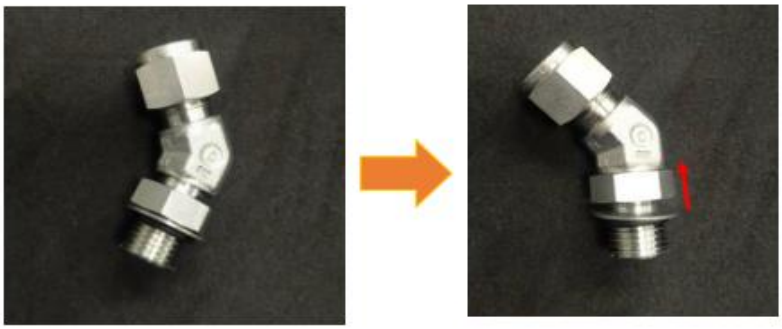
MACHINE: 11/16 open-end wrench 27N·m, 16mm open end

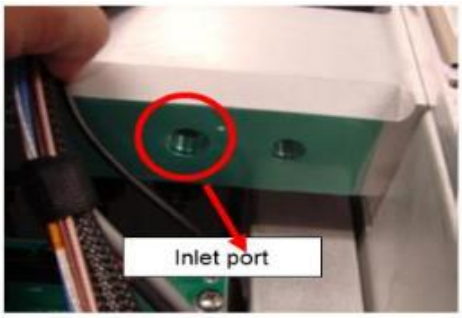
METHOD:

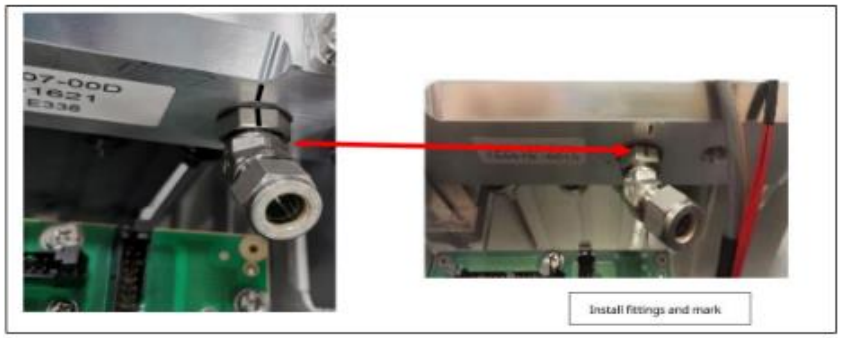
9.2 Install 1pcs 3/8"Pipe connector (361-754-05).

a. Take 1pcs 3/8"Pipe joints (361-754-05), and use Molykote grease mixture (638-030-00) to apply threads and O-rings before installation

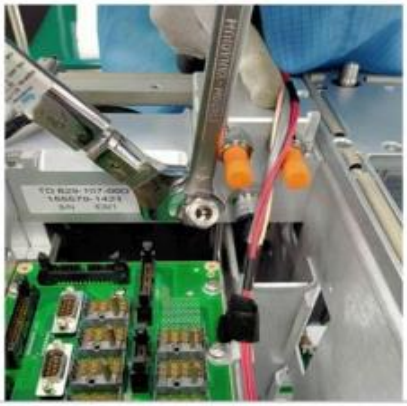
b. Turn the nut back to the direction of the washer to the end.







Tighten the elbow manually and turn it in the opposite direction to correct the position of the elbow pipe joint as in the picture.
(Make sure the elbow position is the same as the picture)



Use 16mm open end to hold the fitting body steady

Then use an open-end wrench and a torque value of 27 N·m to tighten the pipe joint.
After locking the pipe joint, mark with a marker pen.

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MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

Process Aleart: Test Head (Preassure Sensor Assembly)

1. Visual Check O-ring. Ensure No Damaged, No Dented, No Cracked or No Deformation.
2. Ensure installing the O-ring on the pressure sensor and place the O-ring on the bottom of the screw thread (groove).

ACCEPT

Figure 1: Good O-ring that has round shape, smooth surface, No dented, No cracked or No deformation.

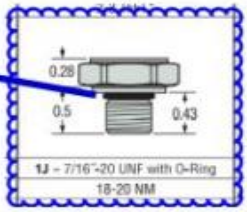

Figure 2: Good O-ring sit on the bottom of the screw thread (groove).

REJECT

Figure 3: Reject O-ring that has damaged, dented, cracked or deformation.

flex

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MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

Process Aleart: Test Head (Preassure Sensor Assembly)

3. Visual Check that the pressure sensor port are not damaged, not broken and no debris.

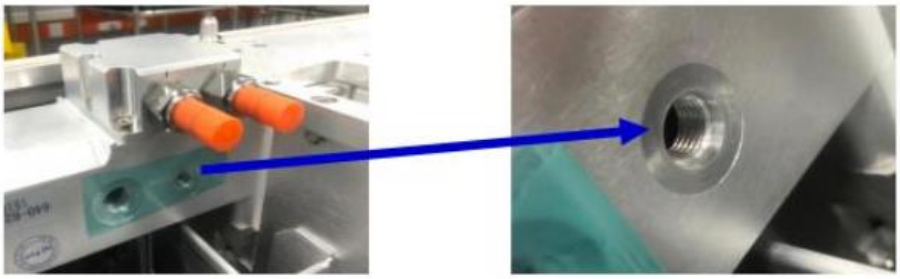


Figure 4: Pressure sensor port in good condition. No damage, No broken, No debris

4. Visual Check the O-ring is flush against the pressure sensor port.

ACCEPT

Figure 5: Good O-ring is flush against the pressure sensor port

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
18	638-079-01	ELEC-MECH, UFII SMALL MIDPLANE ASSY, FOR SEMI TEST	2
19	361-607-08	SCRW PAN HEAD TORX M4X.7P/10 MM SS W/NYLOK	12

MACHINE: 22mm long socket wrench 19 N.m.

METHOD:

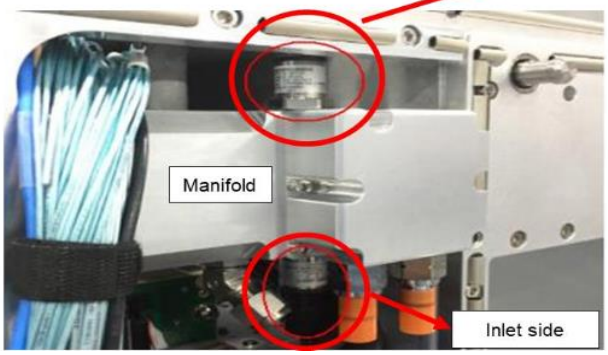
9.3 Install 2 pressure sensors (638-079-01).

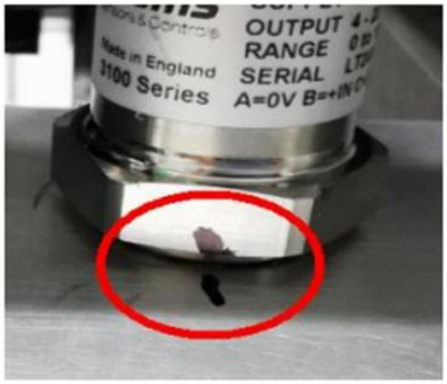
a. Use Molykote grease mixture (638-030-00) to coat the threads of 2 pressure sensors and 2 O-rings (361-607-08).

b. Install the O-ring on the pressure sensor and place the O-ring on the bottom of the thread.

c. Install the pressure sensor at the inlet and outlet of Manifold.

d. Use a 22mm long socket wrench to fix the pressure sensor with a torque value of 19 N·m.





After locking, make a mark with a marker pen on the root of the two sensors.

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: Hex 3mm with 1.4 N·m.

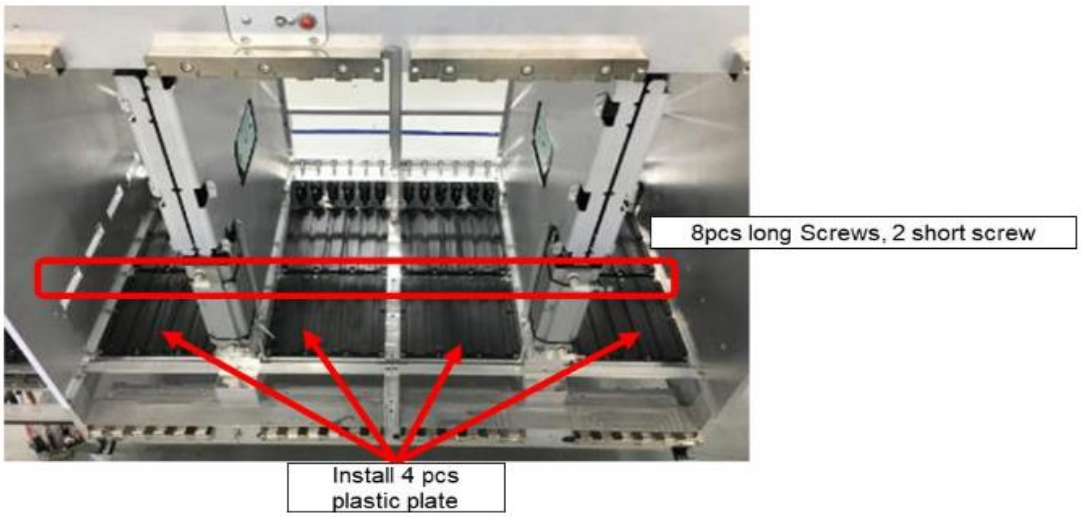
METHOD:

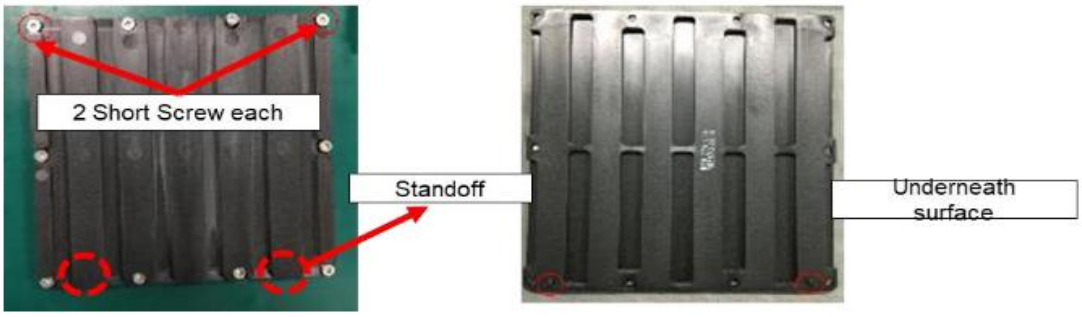
10.1 Rotate the T/H as shown in the figure.

10.2 Reinstall 4 plastic plates:

a. First fix 8 long screws, 2 short screws for each plastic plate, the tool is a Hex 3mm Torx wrench, the torque value is 1.4 N·m.

b. Fix 32 long screws and 8 short screws for each plastic plate. The tool is a Hex 3mm Torx wrench with a torque value of 1.4 N·m.





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MAN: CERTIFIED OPERATOR, TECHNICIAN

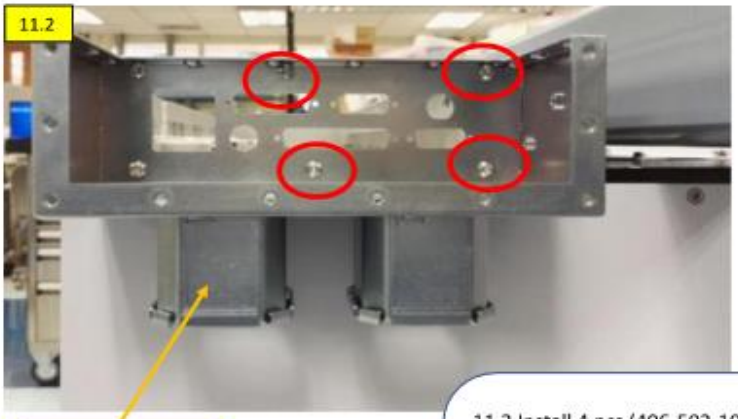
MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
19	638-309-02	ALUMINUM, PANEL, DISCONNECT, BRACKET OR MISC ARTICLE	1
20	406-502-10	SCRW PAN HEAD TORX M5X.8P/10MM SS W/NYLOK	6

MACHINE: 22mm long socket wrench 19 N.m.

METHOD:

11.1 Rotate the T/H direction as shown.



11.2 Install 4 pcs (406-502-10) at the red cirlce location but do not torque yet.

11.3 Leave the remaining 2 holes ash shown in diagram.

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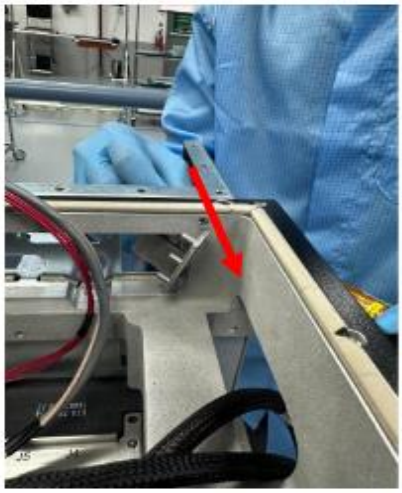
MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
21	692-741-00	MECHANICAL, SHT MENTAL, L BRACKET, FOR SEMI TEST	1

MACHINE: T25 Torx wrench 3.3 N·m.

METHOD:



11.4 Install the Ground Bracket (692-741-00) at the location shown with 2 pcs screw (406-502-10).
11.5 Install the Ground Bracket (692-741-00) in the location shown with 2 screws (406-502-10).

11.6 Torque all 6pcs screw with T25 Bit and 3.3Nm Torque

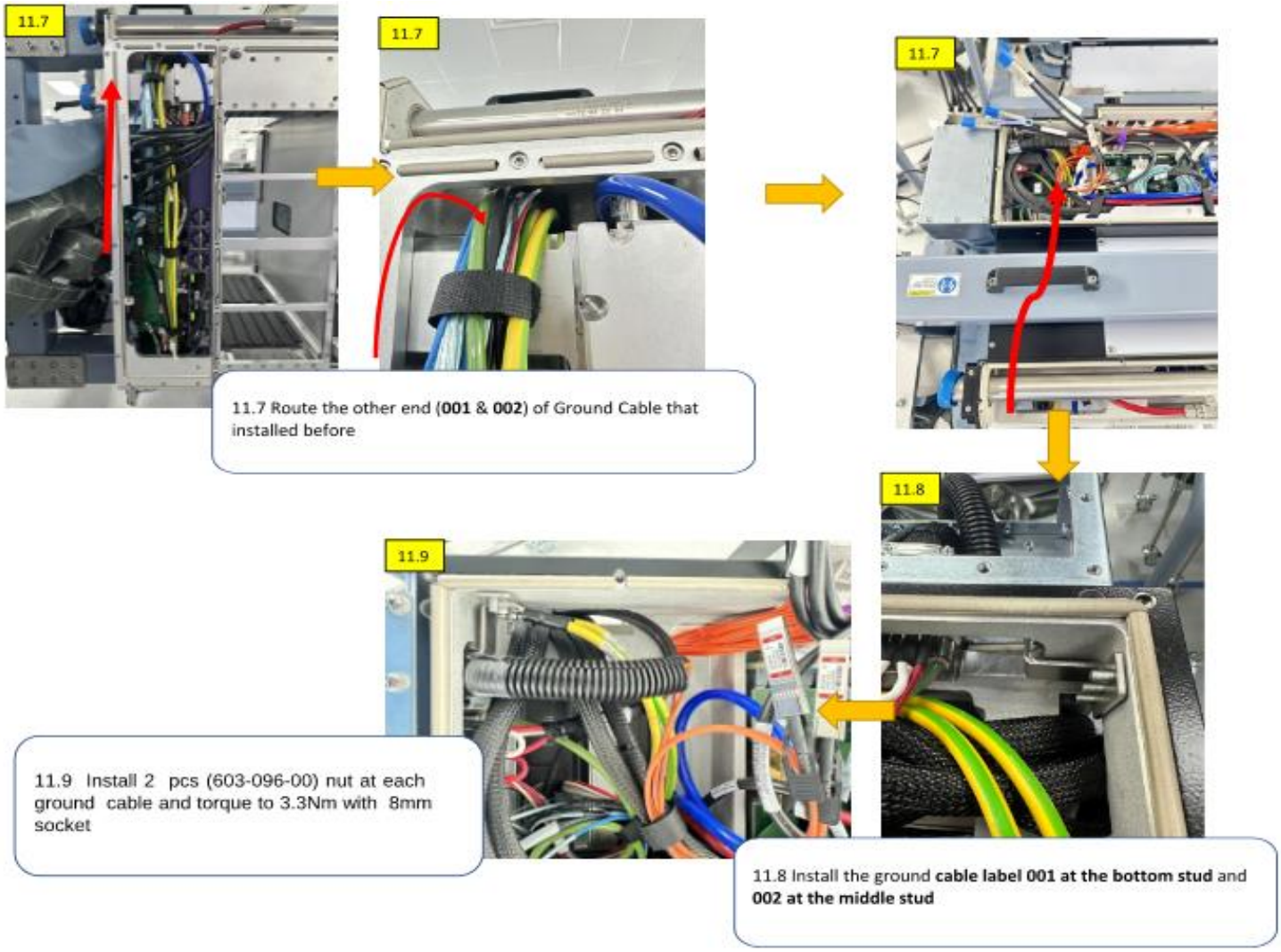
MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
22	603-096-00	NUT HEX SS M5X0.8 W/WSHR W/TTH 8MM WDH	2

MACHINE: 8mm socket, 3.3Nm Torque

METHOD:



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MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: 3/16 socket wrench 0.19 N·m

METHOD:

11.3 Install the EMO cable (652-712-00) and SB24 cable (650-705-00)on the panel as shown in the figure.

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

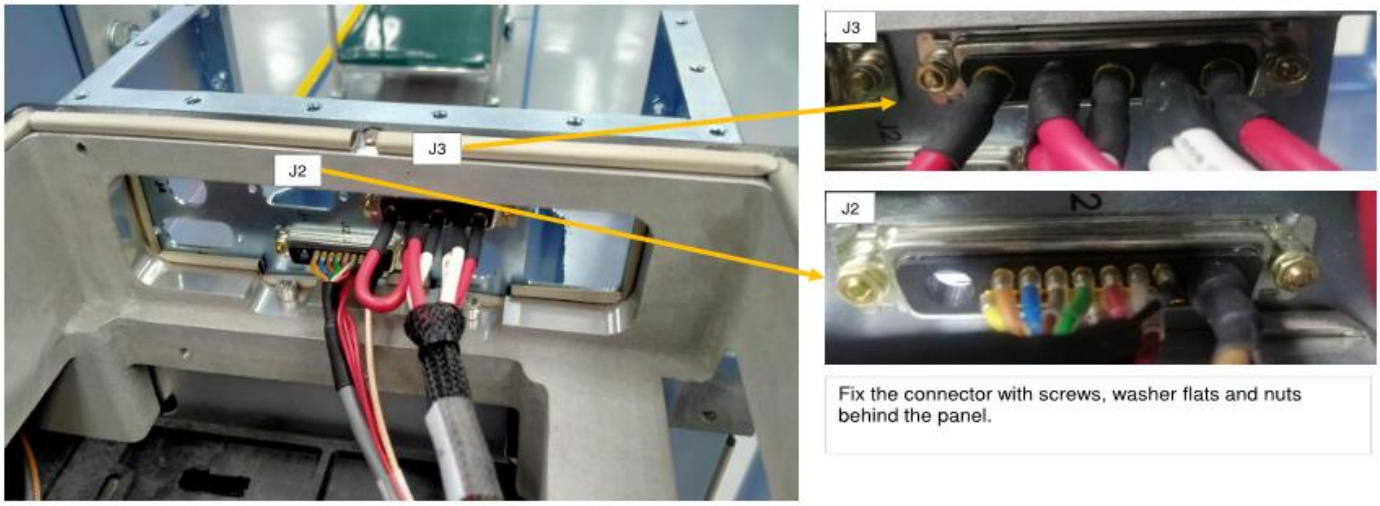
MACHINE: 3/16 socket wrench 0.19 N·m

METHOD: 11.4

a. Insert the wire labeled "T/H DISCON_PNL J3" into the interface labeled "J3" on the panel.

b. Insert the cable labeled "T/H DISCON_PNL J2" into the interface marked "J2" on the panel.

c. Use a tool 3/16 socket wrench with a torque value of 0.19 N·m.



Fix the connector with screws, washer flats and nuts behind the panel.

Front View : No washer